

Revised Generation Interconnection System Impact Study Report

For Queue Project Q21-1868

Mickens 138 kV
September 28, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1868. The Interconnection Customer proposes a 124.0 MW solar photovoltaic facility interconnecting at the Mickens 138 kV substation in the Virginia area. The request was received on August 25, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 642 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 13 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,526,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: WHEELABRATOR PORTSMOUTH / GOSPORT 138kV
Contingency: MICKENS / SWAN 138kV
Loading: 122.0% (Rating: 371 MVA)
DFAX: 14.2%
Pre-project loading: 117.2%
2. Facility: LAKESIDE / NATCHEZ 138kV
Contingency: MITCHELL DAM / SKAGGS ISLAND CT 2 SUBSTATION 345kV
Loading: 106.0% (Rating: 634 MVA)
DFAX: 6.2%
Pre-project loading: 104.8%
3. Facility: SKAGGS ISLAND JUNCTION 2 / MITCHELL DAM 345kV L
Contingency: BEAR / MECKLENBURG POWER STA 345kV
Loading: 105.2% (Rating: 2,032 MVA)
DFAX: 24.5%
Pre-project loading: 103.7%
4. Facility: HOSPITAL DISTRITO 7003 / FIESTA 138kV
Contingency: MITCHELL DAM / SKAGGS ISLAND JCT 2 345kV L
Loading: 104.1% (Rating: 240 MVA)
DFAX: 5.0%
Pre-project loading: 101.5%
5. Facility: SYL ASKIN / ELKO 138kV line
Contingency: HAGGIN / MITCHELL DAM 345kV
Loading: 104.0% (Rating: 378 MVA)
DFAX: 7.0%
Pre-project loading: 101.7%

6. Facility: FRITCH / LAKESIDE 138kV
Contingency: MITCHELL DAM / SKAGGS ISLAND JCT 2 345kV
Loading: 103.9% (Rating: 715 MVA)
DFAX: 6.2%
Pre-project loading: 102.8%
7. Facility: SPRING ST / CHIPOLA 138kV
Contingency: MITCHELL DAM / SKAGGS ISLAND JUNCTION 2 345kV
Loading: 103.1% (Rating: 379 MVA)
DFAX: 7.0%
Pre-project loading: 100.8%
8. Facility: FIESTA / LANEXA 138kV
Contingency: MITCHELL DAM / SKAGGS ISLAND JCT 2 345kV L
Loading: 102.1% (Rating: 228 MVA)
DFAX: 5.0%
Pre-project loading: 99.4%
9. Facility: HARDY REIT / HOSPITAL DISTRITO 7003 138kV
Contingency: SKAGGS ISLAND JCT 2 / MICHELL DAM 345kV line
Loading: 101.6% (Rating: 280 MVA)
DFAX: 5.0%
Pre-project loading: 99.4%
10. Facility: WING LANE / SPRING STREET 138kV
Contingency: MITCHELL DAM / SKAGGS ISLAND JCT 2 345kV L
Loading: 101.3% (Rating: 427 MVA)
DFAX: 7.0%
Pre-project loading: 99.3%
11. Facility: CHIPOLA / WESTCOAST CELLUFIBRE 138kV
Contingency: MITCHELL DAM SUBSTATION / SKAGGS ISLAND JCT 2 345kV
Loading: 100.9% (Rating: 520 MVA)
DFAX: 9.5%
Pre-project loading: 98.6%
12. Facility: BEAR ISLAND SUBSTATION / DUNN SIDING SUBSTATION 345kV L
Contingency: LEAVITT / BIRCHWOD 345kV line
Loading: 100.4% (Rating: 3,342 MVA)
DFAX: 24.4%
Pre-project loading: 99.5%
13. Facility: FRITCH / BIG HANAFORD 138kV
Contingency: SKAGGS ISLAND JCT 2 / MITCHELL DAM 345kV
Loading: 100.3% (Rating: 1,028 MVA)
DFAX: 8.7%
Pre-project loading: 99.2%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
WHEELABRATOR PORTSMOUTH / GOSPORT 138kV	Rebuild line with higher-capacity conductor	\$2,025,000	7% (\$142,000)
LAKESIDE / NATCHEZ 138kV	Reconductor line	\$1,038,000	1% (\$10,000)
MITCHELL DM / SKAGGS ISLAND JUNCTION 2 345kV	Reconductor line	\$26,854,000	1% (\$346,000)
HOSPITAL DISTRITO 7003 / FIESTA 138kV	Reconductor line	\$8,228,000	3% (\$259,000)
SYL LASKIN / ELKO 138kV Line	Reconductor line	\$3,763,000	1% (\$52,000)
FRITCH / LAKESIDE 138kV	Reconductor line	\$862,000	1% (\$8,000)
SPRING STREET / CHIPOLA 138kV	Reconductor line	\$4,639,000	2% (\$75,000)
FIESTA SUBSTATION / LANEXA 138kV	Reconductor line	\$1,889,000	3% (\$60,000)
HARDY REIT / HOSPITAL DISTRITO 7003 138kV L	Reconductor line	\$6,484,000	3% (\$204,000)
WING LANE / SPRING ST 138kV L	Reconductor line	\$995,000	2% (\$16,000)
WESTCOAST CELLUFIBRE / CHIPOLA 138kV L	Reconductor line	\$975,000	1% (\$11,000)
DUNN SIDING / BEAR ISLAND 345kV	Reconductor line	\$27,566,000	1% (\$328,000)
BIG HANAFORD / FRITCH 138kV	Reconductor line	\$1,809,000	1% (\$15,000)
	Total allocated to project		\$1,526,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.